Electronics Engineer Build Guide

Smart Medication Dispenser

SMD-100 Home Device • SMD-200 Travel Device

VERSION	2.0
DATE	February 2026
CLASSIFICATION	Confidential — Engineering Team
SOURCE	Technical Documentation v4.0
GPIO AUTHORITY	Firmware `pins.h`
COMPLIANCE	CE Class IIa, Swissmedic
LOCATION	Lausanne, Switzerland
TOTAL BUILD TIME	~8 hours (SMD-100)

STEP-BY-STEP ASSEMBLY • TESTING • CALIBRATION

Table of Contents

- 0 Prototyping Roadmap
- 1 Workspace Setup & Equipment
- 2 Component Procurement
- 3 PCB Design & Ordering
- 4 PCB Assembly Strategy
- 5 Power Section Build & Verify
- 6 ESP32-S3 MCU Assembly
- 7 Display Connection
- 8 Audio Section
- 9 Motor Driver Section
- 10 Sensor Section
- 11 SD Card Section
- 12 Mechanical Sub-Assembly
- 13 Battery & BMS
- 14 Final Assembly & Enclosure
- 15 Firmware Flashing
- 16 Testing & Calibration
- 17 SMD-200 Travel Device
- 18 Production & Quality Sign-Off
- A Master GPIO Reference
- B Power Budget
- C Common Pitfalls
- D Troubleshooting
- E Second-Source Strategy

PHASE 0

Before You Start — Prototyping Roadmap

Planning Phase

Do NOT jump straight to a custom PCB. Follow this phased approach to save time and money.

PHASE	DURATION	GOAL	DELIVERABLE
1. Breadboard	2 weeks	Prove concept	Working motor + sensor on breadboard
2. Dev Boards	4 weeks	Firmware dev	Complete software on dev kits
3. Custom PCB v1	4 weeks	Integration test	First PCB prototype
4. Custom PCB v2	2 weeks	Fix issues	Production-ready PCB
5. Pilot Build	4 weeks	DFM validation	50 units
6. Production	Ongoing	Scale	Mass production

Recommended Dev Boards (Phase 1–2)

BOARD	MCU	PRICE	USE FOR
ESP32-S3-DevKitC-1	ESP32-S3	$10	Firmware development, sensor testing
ESP32-S3-LCD-EV-Board	ESP32-S3 + 4.3" LCD	$60	Display/UI development with LVGL
WT32-SC01 Plus	ESP32-S3 + 3.5" LCD	$35	Compact UI testing

Start with the $10 DevKit + breadboard to prove motors, sensors, and API connectivity before ordering custom PCBs.

PHASE 1

Workspace Setup & Equipment

1 day

Step 1.1 — ESD-Safe Workstation

- [ ] Install ESD mat on bench
- [ ] ESD wrist strap connected to ground (wear at ALL times)
- [ ] Ground all equipment
- [ ] Maintain 40–60% RH humidity

Step 1.2 — Essential Equipment

#	TOOL	MODEL	PRICE	PURPOSE
1	Soldering station (temp controlled)	Hakko FX-888D	$120	All soldering
2	Hot air rework station	Quick 861DW	$150	QFN packages (BQ24195, TPS62150, MAX98357A, ESP32 module)
3	Digital multimeter	Fluke 117	$200	Voltage/continuity checks
4	Oscilloscope (50MHz+, 4ch)	Rigol DS1054Z	$350	Signal verification, I2C/SPI/PWM debugging
5	Variable bench power supply	Korad KA3005D	$80	Controlled power during testing (0-30V, 0-5A)
6	USB logic analyzer	Saleae Logic 8	$500	I2C/SPI/UART protocol debugging
7	Stereo microscope	AmScope SM-4T	$250	QFN/FFC solder inspection

Step 1.3 — Hand Tools & Consumables

- [ ] Fine ESD tweezers (for 0402/0603)
- [ ] Flush cutters (lead trimming)
- [ ] Wire strippers (22-30 AWG)
- [ ] Solder wick (for rework)
- [ ] Solder paste (SAC305 lead-free)
- [ ] Flux pen (no-clean flux for rework)
- [ ] IPA 99% + lint-free wipes
- [ ] Kapton tape (heat protection)
- [ ] Heat shrink tubing (2mm-10mm)
- [ ] Calibration weights: **10g, 50g, 100g**

- [ ] Solder wire 0.5mm
- [ ] Copper tape (for EMI shielding)

Step 1.4 — Software Environment

- [ ] Install **ESP-IDF v5.1+** (official Espressif SDK)
  Windows: `https://dl.espressif.com/dl/esp-idf/`
  Or: `git clone -b v5.1.2 --recursive https://github.com/espressif/esp-idf.git`
- [ ] Install **VS Code** with ESP-IDF extension
- [ ] `esptool.py` (comes with ESP-IDF)
- [ ] Serial terminal: PuTTY / minicom / ESP-IDF monitor
- [ ] Obtain firmware binary from firmware team
- [ ] USB-to-UART adapter (CP2102 or CH340) as backup programming method

PHASE 2

Component Procurement & Incoming Inspection

1–3 weeks

Step 2.1 — ICs & Modules ORDER FIRST — LONGEST LEAD TIME

REF	COMPONENT	MANUFACTURER	PART NUMBER	QTY	COST	SUPPLIER
U1	MCU Module	Espressif	ESP32-S3-WROOM-1-N16R8	1	$5.50	DigiKey
U2	Power Path IC	TI	BQ24195RGER	1	$3.50	Mouser
U3	5V Buck Regulator	TI	TPS62150RGTR	1	$1.80	DigiKey
U4	3.3V LDO	Diodes Inc	AP2112K-3.3TRG1	2	$0.50	LCSC
U5-U11	Motor Drivers	TI/ST	ULN2003AN	7	$2.10	LCSC
U7	Audio Amp	Maxim/ADI	MAX98357AETE+	1	$2.50	DigiKey
U8	Load Cell ADC	Avia	HX711	1	$0.80	LCSC
U9	Temp/Humidity	Sensirion	SHT40-AD1B	1	$2.50	DigiKey
U10	Light Sensor	ROHM	BH1750FVI	1	$1.00	LCSC
U11	I/O Expander	Microchip	MCP23017-E/SO	1	$1.50	DigiKey

Step 2.2 — Display, Motors, Sensors

REF	COMPONENT	PART	QTY	COST	SUPPLIER
LCD1	4.3" RGB TFT Display	Waveshare 4.3" RGB (GT911 touch)	1	$28.00	Waveshare
M1-M10	Stepper Motors	28BYJ-48 5V	10	$15.00	AliExpress
M11	Gate Servo	SG90 (TowerPro)	1	$2.00	Amazon
LS1	Speaker	CUI CMS-403288-078 (40mm, 8Ω, 2W)	1	$1.50	DigiKey
Q1-Q10	Optical Pill Sensors	Vishay TCPT1300X01	10	$8.00	DigiKey
Q11-Q20	Hall Effect Sensors	Allegro A3144	10	$3.00	LCSC
PIR1	Motion Sensor	AM312 mini PIR	1	$1.50	AliExpress
SW1	Reed Switch + Magnet	NO reed switch + 10×5mm neodymium	1	$0.30	LCSC
LC1	Load Cell	1kg TAL220	1	$3.00	Amazon

Step 2.3 — Power & Battery

COMPONENT	PART	QTY	COST	NOTES
Battery Cells	Samsung INR18650-25R (2500mAh, 20A)	2	$8.00	Buy genuine only! (IMRbatteries, Nkon.nl)
Battery Holder	Keystone 1042 (2×18650 series)	1	$2.00	Series = 7.4V
BMS Board	2S 7.4V 5A (DW01A + MOSFETs)	1	$1.50	Over/under voltage + short protection
AC Adapter	Mean Well GST25E12-P1J (12V/2A)	1	$12.00	**Swiss Type J plug** (SEV 1011) + EU adapter

Step 2.4 — Passives & Connectors

#	COMPONENT	SPEC	QTY	SUPPLIER
1	Ceramic caps	100nF 0402, 16V, X7R (decoupling)	30	LCSC
2	Ceramic caps	10µF 0805, 16V, X5R (bulk)	15	LCSC
3	Ceramic caps	22µF 0805, 10V, X5R (regulator output)	5	LCSC
4	Ceramic caps	1µF 0603, 16V, X7R	10	LCSC
5	Ceramic caps	0.1µF 0402 (ESP32 EN pin RC)	2	LCSC
6	Resistors	10K 0603 ±5% (pull-ups)	20	LCSC
7	Resistors	4.7K 0603 ±5% (I2C pull-ups)	4	LCSC
8	Resistors	100Ω 0603 ±5% (optical sensor LED current)	10	LCSC
9	Inductors	2.2µH shielded, 1.5A, 4×4mm (TPS62150)	2	LCSC
10	Ferrite bead	600Ω @ 100MHz, 0805, 500mA	2	LCSC
11	N-MOSFET	2N7002 SOT-23 (backlight PWM driver)	1	LCSC
12	TVS diodes	ESD protection for USB, touch, external lines	5	LCSC
13	FFC connector	40-pin, 0.5mm pitch, bottom-contact, ZIF	1	LCSC
14	JST-XH 5-pin	Motor connectors	10	LCSC
15	JST-PH 3-pin	Sensor connectors (2.0mm pitch)	12	LCSC
16	JST-XH 2-pin	Battery, speaker connectors	4	LCSC
17	Barrel jack	5.5×2.1mm, panel mount, center positive	1	LCSC
18	SD card socket	Micro SD push-push type	1	LCSC
19	USB connector	USB-B Mini (programming port)	1	LCSC
20	Pin header	2.54mm 4-pin (UART debug)	1	LCSC
PCB1	Main PCB	4-layer, 150×100mm	1	JLCPCB ($8)
PCB2	Motor PCB	2-layer, 80×60mm (if separate)	1	JLCPCB ($3)
ENC1	Enclosure	Custom ABS injection molded	1	Alibaba ($22)
—	SD Card	SanDisk 16GB micro SD	1	Amazon ($5)
—	Packaging	Box, manual, accessories	1	Alibaba ($6)

Total BOM Cost: ~$159 per unit (prototype qty)

Step 2.5 — Incoming Inspection

- [ ] Verify ALL part numbers match the order (cross-check with BOM above)
- [ ] Visually inspect IC packages for damage (bent/missing pins, cracks)
- [ ] **ESP32-S3:** Confirm module marking reads **"N16R8"** (16MB flash + 8MB PSRAM)

- [ ] **Batteries:** Verify cells are **genuine Samsung** (holographic label, weight ~45g each)
  WARNING: Fake 18650 cells are common. Buy ONLY from reputable sources.
- [ ] **AC Adapter:** Measure output with multimeter: **12V ±5%**, check CE/UL marking
- [ ] Store moisture-sensitive ICs (QFN, BGA) in dry cabinet or sealed bag with desiccant
- [ ] Verify display FFC cable connector type matches your PCB footprint

PHASE 3

PCB Design Rules, Ordering & Inspection

3–10 days

Step 3.1 — PCB Layer Stack (4-Layer, 1.6mm)

```
Layer 1 (Top):    Signal + Components       — 1.0 oz copper
Prepreg:          FR-4, 0.2mm
Layer 2:          GROUND PLANE (solid)       — 1.0 oz copper  *** DO NOT SPLIT ***
Core:             FR-4, 1.0mm
Layer 3:          Power Planes (3.3V/5V/12V) — 1.0 oz copper  (split for voltages)
Prepreg:          FR-4, 0.2mm
Layer 4 (Bottom): Signal + Components        — 1.0 oz copper
```

Step 3.2 — Trace/Space Rules

PARAMETER	MINIMUM	PREFERRED	NOTES
Signal trace width	0.15mm (6mil)	0.2mm	Standard signals
Power trace width	0.3mm (12mil)	0.5mm	3.3V, 5V rails
Motor trace width	0.5mm (20mil)	1.0mm	Handle 500mA per motor
Trace spacing	0.15mm (6mil)	0.2mm	Minimum clearance
Via diameter	0.4mm	0.5mm	Outer diameter
Via drill	0.2mm	0.25mm	Hole diameter
Component spacing	0.2mm	0.3mm	Between pads
Board edge clearance	0.3mm	0.5mm	Keep-out zone

Step 3.3 — EMC Design Rules

RULE	IMPLEMENTATION
Ground plane	Solid Layer 2 — **NO splits** under signal traces
Power plane	Split Layer 3 for different voltages, star-point connection
Decoupling	**100nF per power pin**, place within **3mm** of pin
Bulk capacitors	10µF at power entry, 22µF at regulator outputs
Antenna keep-out	**15mm radius clear**, no ground plane under ESP32 antenna
Crystal traces	Keep <5mm, surround with ground guard ring
Motor isolation	**Separate ground return path**, ferrite bead at power entry to motor section
ESD protection	TVS diodes on ALL external connections (USB, barrel jack, touch)

Step 3.4 — Thermal Design

HEAT SOURCE	DISSIPATION	MITIGATION
ESP32-S3	0.5W	Exposed pad to ground plane, via array underneath
BQ24195	0.8W	**Thermal pad MUST connect to ground** via 3×3 via array (0.3mm, 1mm spacing)
TPS62150	0.2W	Ground pad, adequate copper pour
ULN2003A (each)	0.5W	Thermal vias, copper pour
MAX98357A	0.2W	Ground pad

Step 3.5 — Order PCBs

Main PCB (150mm × 100mm, 4-layer):

- [ ] Send Gerber files to JLCPCB (prototype, ~$8) or Eurocircuits (production, ~$80)
- [ ] Specify: 4-layer, 1.6mm, 1oz copper all layers, FR-4, ENIG finish
- [ ] Order minimum **5 pcs** (expect 1-2 for rework during bring-up)

Step 3.6 — PCB Incoming Inspection

- [ ] Visual inspection: no scratches, copper shorts, broken traces
- [ ] Check drill holes align with pad centers
- [ ] Verify soldermask openings are correct (especially QFN thermal pads)
- [ ] Verify FFC connector footprint matches your 40-pin 0.5mm connector
- [ ] Measure board dimensions with calipers (150.0 × 100.0mm ±0.1mm)

- [ ] Continuity test: **3.3V plane NOT shorted to GND**, **5V plane NOT shorted to GND**

PHASE 4

SMD-100 PCB Assembly Strategy

~2 hours total

Assembly Order (Smallest to Largest)

ORDER	COMPONENT TYPE	REFERENCE	METHOD
1	0402 capacitors (100nF decoupling)	C1-C30	Solder paste + hot air
2	0603 resistors (10K, 4.7K, 100Ω)	R1-R50	Solder paste + hot air
3	0805 capacitors (10µF, 22µF bulk)	C31-C45	Solder paste + hot air
4	SOT-23/SOT-25 (2N7002, AP2112K)	Q1-Q5, U4	Hand solder or hot air
5	QFN-16 (TPS62150, 3×3mm)	U3	**Hot air**, microscope alignment
6	QFN-24 (BQ24195, 4×4mm)	U2	**Hot air**, thermal pad critical
7	QFN-16 (MAX98357A, 3×3mm)	U7	**Hot air**, microscope alignment
8	ESP32-S3 module (25.5×18mm)	U1	**Preheat + hot air** reflow
9	SOIC packages (ULN2003A ×7, MCP23017, HX711)	U5-U11	Hand solder or reflow
10	Through-hole connectors	J1-J20	Hand solder
11	FFC connector (40-pin 0.5mm)	J_LCD	Fine tip hand solder
12	SD card socket	J_SD	Hand solder

PHASE 5

Power Section — Build & Verify

45 min assembly + 15 min testing

CRITICAL: Build and test power FIRST, before installing ESP32. If there's a voltage issue, you'll fry the $5.50 MCU.

Step 5.1 — BQ24195 Power Path Controller QFN-24, 4×4MM

1. Apply solder paste to all 24 QFN pads **+ center thermal pad**
2. **Thermal pad is critical** — this is the heat sink. Must have solder and thermal vias beneath
3. Place BQ24195RGER with tweezers, align pin 1 dot marking
4. Verify alignment under microscope
5. Hot air reflow: **260°C peak, 30 seconds**, nozzle 2-3mm above
6. IC will "settle" when solder melts — watch edges
7. **Inspect ALL 24 pins under microscope** — no bridges, no opens
8. Install passives: **10µF** (VIN, within 3mm), **22µF** (SYS & BAT output), NTC resistor (10K for battery temp)

Step 5.2 — TPS62150 Buck Regulator QFN-16, 3×3MM → 5V Rail

1. Apply solder paste to QFN-16 pads + ground pad
2. Place TPS62150RGTR, align pin 1
3. Hot air reflow at 260°C, 20-30 seconds
4. Install **2.2µH shielded inductor** (L1) — place close, keep trace short
5. Install input cap: **10µF** close to VIN
6. Install output cap: **22µF** close to VOUT
7. If adjustable output: install feedback resistor divider for 5.0V

Step 5.3 — AP2112K-3.3 LDO (SOT-25) → 3.3V Rail

1. Hand solder the 5-pin SOT-25 package
2. Install input cap: **1µF** on input pin
3. Install output cap: **10µF** on output pin
4. Install **ferrite bead** (600Ω @ 100MHz) on 3.3V input from 5V rail

Step 5.4 — Power Connectors

1. Solder 12V **barrel jack** (5.5×2.1mm, center positive)
2. Solder **battery connector** (JST-XH 2-pin)
3. Install power input protection (TVS diode for overvoltage, Schottky for reverse polarity if designed)

Step 5.5 — POWER TEST (Critical Gate — DO NOT PROCEED until ALL pass)

If ANY test fails: STOP. Check solder joints under microscope, verify component orientation (pin 1), check for bridges. Fix before proceeding.

#	TEST	HOW	EXPECTED	MEASURED	PASS?
1	Apply 12V via bench supply	Set PSU to 12.0V, limit 500mA	No smoke, no excess current	----------	☐
2	SYS output voltage	Probe BQ24195 SYS pin	~12V	----------	☐
3	**5V rail**	Probe TPS62150 output	**5.0V ±2%**	----------	☐
4	**3.3V rail**	Probe AP2112K output	**3.3V ±2%**	----------	☐
5	Quiescent current	Read bench PSU current	**<10mA** (no load)	----------	☐
6	Heat check	Touch ICs after 30s	No excessive heat	—	☐
7	Short check	Measure 3.3V to GND resistance	**>1KΩ**	----------	☐

PHASE 6

ESP32-S3 MCU Assembly & First Boot

30 minutes

Step 6.1 — Pre-Installation

- [ ] Verify 3.3V rail is stable at 3.3V ±2% (from Phase 5)
- [ ] Verify no shorts between 3.3V and GND at ESP32 pad locations
- [ ] Clean PCB pads with IPA

Step 6.2 — Solder ESP32-S3-WROOM-1-N16R8

1. Apply **flux** generously to all module pads on PCB
2. Apply thin layer of solder paste (stencil method preferred)
3. Place module carefully with tweezers
4. **Check alignment under microscope** — pin 1 marker must match
5. **Preheat board** to 150°C on hot plate (prevents thermal shock)
6. Hot air reflow: focus on module at **260°C for 20-30 seconds**
7. Watch corner pins — solder flows there first
8. Let cool **naturally** — do NOT blow cold air
9. **Inspect ALL pins under microscope** — look for bridges and cold joints
10. Touch up bridges or opens with fine tip iron + flux

Step 6.3 — Critical Support Components

- [ ] **EN pin:** 10K pull-up to 3.3V **+** 0.1μF cap to GND (RC delay for clean boot)
- [ ] **GPIO0 (BOOT):** 10K pull-up to 3.3V + tactile button to GND (for download mode)
- [ ] **Reset button:** Tactile button between EN pin and GND
- [ ] **Decoupling:** Place **10μF + 0.1μF** within 3mm of EACH VCC pin on module
- [ ] **USB pins:** GPIO20 (USB_D-) and GPIO21 (USB_D+) to USB-B Mini connector or pin header

Step 6.4 — Antenna Clearance Verification

> **15mm clearance** around antenna area (top-right of module). **No ground plane** under antenna on ANY layer. **No traces** crossing antenna feed line. No metal enclosure parts within 10mm.

Step 6.5 — ESP32 First Boot Test

1. Connect USB cable (or USB-UART adapter to TX0/RX0 pins)
2. Open serial terminal: **115200 baud, 8N1**
3. Apply 12V power → ESP32 should boot, expect "rst:0x1 (POWERON_RESET)"
4. If **no output:** Hold BOOT → press RESET → release BOOT (enters download mode). Check EN=3.3V, GPIO0=HIGH
5. Run `esptool.py --port COMx flash_id` → should report **16MB flash**
6. If PSRAM test firmware available: boot log shows **"8MB PSRAM initialized"**

PHASE 7

Display Connection & Test

30 minutes

Two separate I2C buses:
- **I2C Bus 0 (Touch):** GPIO38 (SDA) + GPIO39 (SCL) — GT911 at 0x5D
- **I2C Bus 1 (Sensors):** GPIO44 (SDA) + GPIO46 (SCL) — SHT40, BH1750, MCP23017

Step 7.1 — Solder FFC Connector

1. Solder 40-pin, 0.5mm pitch FFC connector (bottom-contact type, ZIF latch)
2. Use fine tip iron at 300°C with flux
3. **Inspect EVERY pin under microscope** — bridges here = dead display

Step 7.2 — Install Backlight Driver

1. Solder N-MOSFET (2N7002, SOT-23): **Gate** → GPIO42, **Drain** → backlight, **Source** → GND
2. Install TVS diodes on touch I2C lines (ESD protection from user touching screen)

Step 7.3 — Display Wiring (from firmware `pins.h`)

GPIO	SIGNAL	GPIO	SIGNAL	GPIO	SIGNAL
1	LCD_R0	6	LCD_G0	12	LCD_B0
2	LCD_R1	7	LCD_G1	13	LCD_B1
3	LCD_R2	8	LCD_G2	14	LCD_B2
4	LCD_R3	9	LCD_G3	15	LCD_B3
5	LCD_R4	10	LCD_G4	16	LCD_B4
—	—	11	LCD_G5	—	—
45	**LCD_CLK**	**48**	**LCD_DE**	**47**	**LCD_VSYNC**
21	**LCD_HSYNC**	42	Backlight PWM	—	—
38	TOUCH_SDA	39	TOUCH_SCL	40	TOUCH_INT
41	TOUCH_RST	—	—	—	—

Step 7.4 — Connect & Test Display

1. Insert FFC cable — **verify orientation** (contacts face down for bottom-contact connector)
2. Close ZIF latch firmly
3. Power on → backlight should illuminate (GPIO42 HIGH)
4. Flash LVGL test firmware
5. **Verify:** All pixels working, correct colors (no swapped R/G/B), no dead lines
6. **Touch test:** Tap all 4 corners + center → verify correct coordinate mapping
7. I2C scan should find GT911 at address **0x5D**

PHASE 8

Audio Section

20 minutes

Step 8.1 — Install MAX98357A (QFN-16, 3×3mm)

1. Apply solder paste to QFN-16 pads
2. Place with tweezers, align under microscope
3. Hot air reflow at 260°C, 20 seconds
4. Install: **10µF** cap on VDD (bulk) + **100nF** cap on VDD (close to chip)

Step 8.2 — I2S Audio Wiring (from firmware `pins.h`)

ESP32 GPIO	MAX98357A PIN	FUNCTION
GPIO17	BCLK	I2S bit clock
GPIO18	LRCLK	I2S left/right clock
GPIO15	DIN	I2S audio data out
—	GAIN → GND	Sets 15dB gain
—	SD → 3.3V	Enable (always on)
—	VDD → 5V	Power supply
—	GND → GND	Ground

Note: Firmware `pins.h` defines GPIO15 for I2S_DOUT, but GPIO15 is also LCD_B3. If using RGB565 (16-bit), fewer blue pins are needed and GPIO15 is freed for audio. **Verify your PCB schematic resolves this conflict.**

Step 8.3 — Speaker Connection

1. Solder 2-pin JST or screw terminal
2. MAX98357A OUT+ → Speaker +, OUT- → Speaker -
3. Speaker: **40mm, 8Ω, 2W** (no polarity for dynamic speaker)

Step 8.4 — Audio Test

- [ ] Flash test firmware with 1kHz tone generator
- [ ] Connect speaker → should hear **clean 1kHz sine tone**
- [ ] Verify volume control works (10 levels via software)
- [ ] Check for distortion at max volume → should be clean at 8Ω load

PHASE 9

Motor Driver Section

30 minutes

Step 9.1 — Install MCP23017 I/O Expander (SOIC-28)

1. Solder MCP23017-E/SO
2. Tie address pins: **A0=GND, A1=GND, A2=GND** → I2C address = **0x20**
3. Install 100nF decoupling cap on VDD pin
4. Connect to **I2C Bus 1:** SDA → GPIO44, SCL → GPIO46
5. **4.7K pull-up resistors** on SDA and SCL (if not already installed)
6. **Quick test:** I2C scan → should detect device at **0x20**

Step 9.2 — Install ULN2003A Motor Drivers (×7)

1. Solder all 7× ULN2003AN (SOIC-16 or DIP-16)
2. For **each** ULN2003A: **COM pin** → **5V** (critical — built-in flyback diodes need this)
3. GND pin → ground
4. Input pins ← MCP23017 GPIO outputs (GPA0-7, GPB0-7)
5. Output pins → motor connector pins

Step 9.3 — Motor Connectors

1. Solder 10× JST-XH 5-pin connectors
2. Wiring per motor (28BYJ-48): Pin 1=**Red** (5V common), Pin 2=**Blue** (A), Pin 3=**Pink** (B), Pin 4=**Yellow** (C), Pin 5=**Orange** (D)

3. Gate servo connector (3-pin): VCC(5V), GND, Signal → **GPIO43**

Step 9.4 — Motor EMI Mitigation

- [ ] Add **100nF ceramic cap across EACH motor coil** (motor noise causes WiFi interference)
- [ ] Use **ferrite beads** on motor power lines entering the PCB
- [ ] Keep motor traces on separate ground return path from digital signals

Step 9.5 — Motor Driver Test

1. Connect one 28BYJ-48 motor
2. Flash motor test firmware
3. Run half-step sequence (8-step: 2048 steps = 1 revolution)
4. Motor should rotate smoothly in both directions
5. Measure current: **~200mA active, ~0mA when off**
6. Test all 10 motor outputs

PHASE 10

Sensor Section (I2C, Digital, Analog)

45 minutes

Reminder: Sensor I2C Bus 1 = GPIO44 (SDA), GPIO46 (SCL)

Step 10.1 — I2C Sensors

SHT40 Temp/Humidity (address 0x44)

1. Tiny DFN 1.5×1.5mm — requires hot air station
2. Apply solder paste, place with tweezers, reflow at 260°C
3. Add 100nF decoupling cap on VDD
4. Accuracy: ±0.2°C temperature, ±1.8% humidity

BH1750 Light Sensor (address 0x23)

1. Small WSOF package, hot air reflow
2. ADDR pin → GND (sets address to 0x23)
3. Add 100nF decoupling cap
4. Range: 1-65535 lux, 1 lux resolution

Step 10.2 — Load Cell + HX711 ADC

1. Solder HX711 module (or SOP-16 IC directly on PCB)
2. HX711 DOUT → **GPIO32**, HX711 SCK → **GPIO33**
3. Load cell: **Red**(E+)→HX711 E+, **Black**(E-)→HX711 E-, **White**(A-)→HX711 A-, **Green**(A+)→HX711 A+
4. HX711 power: VCC → 5V, GND → GND
5. **Place HX711 physically close to load cell** (short wires = less noise)
6. **Shield from motor EMI** (copper tape, physical distance, or ground guard ring)

Step 10.3 — Digital Sensors

PIR Motion Sensor (AM312)

1. Solder 3-pin JST-PH connector (VCC, GND, OUT)
2. Signal output → **GPIO26**. No pull-up needed (push-pull output)
3. 3.3V compatible, 20µA standby. Detects motion at 3-5m, 100° angle

Reed Switch (Door/Tray sensor)

1. Solder 2-pin JST connector
2. One wire → **GPIO27** + **10K pull-up to 3.3V**. Other wire → GND
3. Mount magnet (10×5mm neodymium) on moving part
4. Logic: Door **closed** (magnet near) = GPIO **LOW** / Door **open** = GPIO **HIGH**

Step 10.4 — Optical Pill Sensors (×10)

Each TCPT1300X01 sensor circuit:

```
5V ──[100R]──►IR LED──► GND     (LED side, current ~30mA)
5V ──[10K pull-up]──┬──► MCP23017 GPIO  (phototransistor collector)
                    └──►Phototransistor──► GND (emitter)
```

- [ ] Solder 10× 3-pin connectors (VCC, GND, Signal)
- [ ] Route 8 sensors → MCP23017 Port B (GPB0-GPB7) with interrupt
- [ ] Route 2 remaining → ESP32 GPIO28, GPIO29 (direct)
- [ ] Logic: **No pill = GPIO HIGH** (light passes) / **Pill passing = GPIO LOW** (blocked)
- [ ] Count **rising edges** for pill count

Step 10.5 — Hall Effect Sensors (×10, A3144)

- [ ] 10× A3144 sensors (one per motor position for carousel homing)
- [ ] Each sensor: VCC (5V), GND, Output → **10K pull-up to 3.3V** → MCP23017 GPIO
- [ ] Mount near carousel gear teeth, magnet on carousel at home position

Step 10.6 — I2C Bus Verification

Run I2C scan from ESP32:

BUS 1 (GPIO44/46)		
0x20	MCP23017	☐ Found
0x23	BH1750	☐ Found
0x44	SHT40	☐ Found

BUS 0 (GPIO38/39)		
0x5D	GT911 (Touch)	☐ Found

PHASE 11

SD Card Section

10 minutes

Step 11.1 — Install SD Card Socket

1. Solder micro SD push-push socket
2. SPI wiring: **GPIO36**→SD_CLK, **GPIO35**→SD_CMD (MOSI), **GPIO37**→SD_DATA0 (MISO)
3. Add **10K pull-up resistors** on all SD data lines (required by SD spec)
4. VCC → 3.3V, GND → GND. Add 100nF decoupling cap on VCC

Step 11.2 — SD Card Test

- ☐ Insert 16GB SanDisk micro SD card (formatted FAT32)
- ☐ Flash test firmware that writes/reads a file
- ☐ Verify: write speed >1MB/s, read speed >2MB/s
- ☐ SD card stores: audio files (~5MB for 4 languages), event logs, calibration data

PHASE 12

Motor & Mechanical Sub-Assembly

1 hour

Step 12.1 — Mount Stepper Motors

1. Mount 10× 28BYJ-48 motors to bracket (3D printed or CNC aluminum)
2. Use M3 screws (2 per motor). All shafts same direction
3. Verify each motor rotates freely (no binding)

Step 12.2 — Carousel Assembly

1. Press-fit gear onto each motor's D-shaped shaft
2. Install central carousel with 10 pill compartments
3. Ensure carousel meshes smoothly with motor gears
4. Test full 360° rotation by hand — must be smooth, no grinding

Step 12.3 — Gate Servo (SG90)

1. Mount SG90 in gate mechanism
2. Connect: VCC(5V), GND, Signal → **GPIO43**
3. PWM: 50Hz, **1ms pulse = 0° (closed)**, **2ms pulse = 180° (open)**
4. Test full open/close — gate must fully seal

Step 12.4 — Optical Sensors in Pill Chutes

- [ ] Mount one TCPT1300X01 in each of 10 pill chutes (3mm sensor gap)
- [ ] IR LED on one side, phototransistor opposite
- [ ] Wire each sensor to its JST connector on main PCB

Step 12.5 — Hall Sensors for Homing

- [ ] Mount one A3144 near each motor's gear position

- [ ] Glue small neodymium magnet to carousel at home position for each slot
- [ ] Hall sensor triggers = motor has reached the "home" position

Step 12.6 — Load Cell in Output Tray

- [ ] Fix one end of TAL220 1kg load cell rigidly to device frame
- [ ] Attach output tray to free/floating end — **no mechanical contact with frame**
- [ ] Wire to HX711 (Red→E+, Black→E-, White→A-, Green→A+)
- [ ] Shield HX711 from motor EMI (copper tape or physical distance)

PHASE 13

Battery & BMS Installation

20 minutes

Configuration: 2S1P (2 cells in series) = **7.4V nominal, 5000mAh**

Step 13.1 — BMS Protection Requirements (DW01A + MOSFETs)

PROTECTION	THRESHOLD
Overvoltage cutoff	8.4V (4.2V per cell)
Undervoltage cutoff	6.0V (3.0V per cell)
Overcurrent cutoff	5A
Short circuit	Instant cutoff
Balance current	50-100mA

Step 13.2 — Installation

1. Insert 2× Samsung INR18650-25R cells into Keystone 1042 holder (series connection)
2. **VERIFY POLARITY** — Cell 1 positive → Cell 2 negative (series = 7.4V)
3. Connect BMS board between battery and output connector
4. Measure pack voltage: should read **7.0-8.4V** depending on charge state
5. Connect JST-XH 2-pin to PCB battery input
6. **Test BQ24195 charging:** Apply 12V, BQ24195 should start charging (monitor via I2C or LED)

Step 13.3 — Battery Safety

NEVER short battery terminals. **NEVER** puncture or deform cells. **NEVER** use screws through or near battery. Secure holder with foam tape or mechanical clip only. Store charged batteries away from flammable materials.

PHASE 14

Final Assembly & Enclosure

1 hour

Step 14.1 — Prepare Enclosure (305 × 203 × 254mm)

- [ ] Install heat-set brass inserts: M3 for PCB, M2.5 for display
- [ ] Install cable clips and routing guides

Step 14.2 — Install Components

ORDER	COMPONENT	METHOD
1	Display	Foam gasket around edge, insert FFC cable (**check orientation!**), clip secure
2	Main PCB	M3 standoffs (10mm), screw 4 corners
3	Motor/Carousel	Mount in position, verify clearance for rotation
4	Speaker	Hot glue or foam mount in speaker cavity
5	Battery pack	Foam tape mount (NO screws near battery)
6	PIR sensor	Mount facing front, detection window clear
7	Reed switch	Mount on tray rail, magnet on tray

Step 14.3 — Final Wiring

1. Connect 10× motor cables to JST connectors
2. Connect 10× optical sensor cables
3. Connect 10× hall sensor cables
4. Connect load cell/HX711 cable
5. Connect PIR sensor cable
6. Connect reed switch cable
7. Connect speaker cable
8. Connect battery pack
9. Connect barrel jack for AC power
10. Insert SD card

Step 14.4 — Pre-Close Inspection

- [ ] **No pinched or stressed wires**
- [ ] **All connectors fully seated** (tug test gently)
- [ ] **Battery polarity correct** (measure voltage at PCB connector)
- [ ] **All moving parts have clearance** (carousel, gates, tray)
- [ ] **FFC cable not under stress** (no sharp bends)
- [ ] **Antenna area clear** of metal parts
- [ ] Close enclosure with screws (do not overtighten ABS — it cracks)

PHASE 15

Firmware Flashing & Configuration

30 minutes

Step 15.1 — Flash Firmware

Windows (ESP-IDF command prompt):

```
cd %USERPROFILE%\esp\smart_dispenser
idf.py set-target esp32s3
idf.py build
idf.py -p COM3 flash monitor
```

Direct flash with esptool:

```
esptool.py --chip esp32s3 --port COM3 --baud 921600 write_flash ^
  0x0 bootloader.bin ^
  0x10000 firmware.bin ^
  0x8000 partition-table.bin
```

Step 15.2 — Verify Boot

Serial monitor at **115200 baud**. Expected boot messages:

- [ ] ESP32-S3 identified
- [ ] 16MB Flash detected
- [ ] 8MB PSRAM initialized
- [ ] FreeRTOS tasks started (main, wifi, ui, dispense, sensor, audio)
- [ ] LVGL initialized, Display driver OK
- [ ] I2C devices found (0x20, 0x23, 0x44, 0x5D)
- [ ] WiFi scanning...

Step 15.3 — Initial Configuration

1. Configure WiFi: via touch screen setup wizard or serial command
2. Verify WiFi connects in **<30 seconds**
3. Program device serial number: **SMD-XXXXXXXX** format (stored in NVS)

4. Register device with cloud API: `POST /api/v1/devices/register`
5. Verify heartbeat received by server (every 60 seconds)

PHASE 16

Full Testing & Calibration

1–2 hours

16.1 — Power System Tests

TEST	EXPECTED	MEASURED	PASS?
12V input, idle power	<5W	_____	☐
5V rail under full load	5.0V ±2%	_____	☐
3.3V rail under full load	3.3V ±2%	_____	☐
AC→Battery switchover time	<100ms (unplug AC)	_____	☐
Battery→AC switchover time	<100ms (plug AC)	_____	☐
Battery backup, idle	>48 hours	_____	☐
Low battery alert	Triggers at 20%	—	☐
Critical shutdown	At 5%	—	☐
Reverse polarity protection	No damage	—	☐

16.2 — Display & Touch Tests

TEST	EXPECTED	PASS?
Display powers on	Screen lit in <2s	☐
Full white screen	All pixels white, no dead pixels	☐
Full black screen	All pixels black	☐
Color bars (R/G/B)	Correct colors, no channel swap	☐
Brightness control	10 PWM levels via GPIO42	☐
Touch: top-left corner	Responds correctly	☐
Touch: top-right corner	Responds correctly	☐
Touch: bottom-left corner	Responds correctly	☐
Touch: bottom-right corner	Responds correctly	☐
Touch: center	Responds correctly	☐
Touch response time	<50ms	☐
Viewing angle	Readable at ±45° (IPS)	☐

16.3 — Motor Tests

TEST	EXPECTED	PASS?
Motor 1: Full CW rotation	Smooth, 2048 steps = 360°	☐
Motor 2: Full CW rotation	Smooth	☐
Motor 3: Full CW rotation	Smooth	☐
Motor 4: Full CW rotation	Smooth	☐
Motor 5: Full CW rotation	Smooth	☐
Motor 6: Full CW rotation	Smooth	☐
Motor 7: Full CW rotation	Smooth	☐
Motor 8: Full CW rotation	Smooth	☐
Motor 9: Full CW rotation	Smooth	☐
Motor 10: Full CW rotation	Smooth	☐
Direction reversal (all)	No lost steps	☐
Gate servo: Open (90°)	Fully opens	☐
Gate servo: Close (0°)	Fully seals	☐
Motor noise level	<45 dB @ 1m	☐
Hall sensor homing (all 10)	Each finds home	☐

16.4 — Sensor Tests

TEST	EXPECTED	PASS?
SHT40: Temperature	±2°C of reference	☐
SHT40: Humidity	±5% RH of reference	☐
BH1750: Light	Changes with ambient	☐
PIR: Motion at 1m	GPIO26 goes HIGH	☐
PIR: No motion 10s	GPIO26 goes LOW	☐
Reed switch: Magnet near	GPIO27 LOW (closed)	☐
Reed switch: Magnet away	GPIO27 HIGH (open)	☐
Optical sensor 1: Pass pill	Correctly counts 1	☐
Optical sensors 2-10	All count correctly	☐
Optical: 5 pills consecutive	Counts exactly 5	☐
SD card: Read/write test	File created & read back	☐

16.5 — Load Cell Calibration CRITICAL

1. Enter calibration: `Settings → Calibration → Load Cell` (or serial: `calibrate loadcell`)
2. Remove everything from output tray → press **"Zero"** → records offset
3. Place **50g calibration weight** → press **"Calibrate"** → records scale factor
4. **Verify:** Display shows **50.0g ±1g**
5. Remove weight, place **10g** → should read **10.0g ±0.5g**
6. Remove weight → should return to **0.0g ±0.1g**
7. Save calibration to NVS

16.6 — Touch Screen Calibration

1. Enter: `Settings → Calibration → Touch`
2. Tap each of 4 corner crosshair targets precisely
3. Tap center target
4. Verify touch accuracy across screen: **<5mm error everywhere**
5. Save calibration

16.7 — Motor Homing Calibration

1. Enter: `Settings → Calibration → Motors`
2. For each motor (1-10): Run **"Find Home"** → motor rotates until Hall sensor triggers
3. Position saved as home (slot 1 position)
4. Verify: command to go to slot N → carousel stops at correct slot every time

16.8 — Full Dispense Integration Test

Run `dispense 1 2` (slot 1, 2 pills) and verify the full sequence:

- [ ] Carousel rotates to slot 1 (~500ms)
- [ ] Hall sensor confirms correct position
- [ ] Gate opens (servo to 90°)
- [ ] Optical sensor counts exactly 2 pills falling

- [ ] Gate closes (servo to 0°)
- [ ] Load cell verifies weight matches expected (2 pills × pill weight)
- [ ] DOSE_DISPENSED event sent to cloud API
- [ ] **Repeat for ALL 10 slots**
- [ ] Test **jam detection:** deliberately block pill chute → should generate Error E101

16.9 — Connectivity Tests

TEST	EXPECTED	PASS?
WiFi connects (2.4GHz)	<30 seconds	☐
WiFi connects (5GHz)	<30 seconds	☐
WiFi auto-reconnect	Reconnects in <60s after AP off/on	☐
BLE advertising	Discoverable by phone	☐
API heartbeat	Every 60s ±5s	☐
Heartbeat data	Battery%, temp, WiFi RSSI, slot data	☐
HTTPS/TLS 1.3	Certificate validated	☐
OTA update	Firmware downloads and installs	☐

16.10 — Audio Tests

TEST	EXPECTED	PASS?
1kHz test tone	Clear, no distortion	☐
Volume: 10 levels	Each level distinguishable, 60-90 dB	☐
Alert tone	Distinct and attention-getting	☐
Voice prompt (if loaded)	Intelligible speech	☐
Quiet hours mode	Audio fully muted	☐

PHASE 17

SMD-200 Travel Device Build

2–3 hours

Step 17.1 — Travel Device BOM Additions

COMPONENT	PART	QTY	PRICE	NOTES
MCU	ESP32-S3-MINI-1	1	$4.00	Smaller module, single-core 160MHz
Cellular	SIM7080G (SIMCom)	1	$12-15	LTE Cat-M1 + GPS, 17.6×15.7mm LGA
Display	SSD1306 2.4" OLED (128×64, I2C)	1	$8.00	Address 0x3C
Charge IC	TP4056 + DW01 protection	1	$0.50	USB-C 5V/1A charging
Fuel Gauge	MAX17048	1	$2.00	I2C battery monitor
Battery	Li-Po 3000mAh pouch (3.7V, JST-PH)	1	$8.00	60×50×8mm
Servos	SG90 micro	4	$1.50 ea	Gate control per slot
Optical	TCPT1300X01	4	$0.80 ea	Pill counting
SIM card	Nano SIM socket (push-push)	1	$0.80	
Antennas	u.FL LTE + u.FL GPS antenna	2	$5.00 ea	Connect LAST
USB-C	16-pin connector	1	$0.80	Charging + data

Step 17.2 — Travel PCB Assembly (80×60mm, 2-layer)

ORDER	COMPONENT	NOTES
1	Passives (0603)	Solder paste + hot air
2	ESP32-S3-MINI-1	Hot air reflow
3	SIM7080G	**Large LGA (17.6×15.7mm)** — hot air at 270°C
4	TP4056 module	SOT-23-6 or breakout module
5	MAX17048 fuel gauge	Tiny package, hot air
6	Connectors	USB-C, SIM socket, u.FL — hand solder

Step 17.3 — SIM7080G Installation (Most Challenging)

1. Apply solder paste to ALL pads + large ground paddle
2. Place with tweezers (17.6 × 15.7mm)
3. Hot air at **270°C** from above — watch for solder to flow at edges
4. Module "settles" when solder melts
5. **Inspect under microscope** for bridges (especially ground paddle area)
6. Attach u.FL antenna connectors **LAST** (cables are fragile)

Step 17.4 — Travel Device GPIO Map (Section 11.2 — authoritative)

GPIO	FUNCTION	DIR	NOTES
0	BOOT button	In	Pull-up
1	I2C SDA	Bidir	Display (SSD1306) + sensors
2	I2C SCL	Out	Shared bus
3	**Cellular TXD**	Out	→ SIM7080G RXD
4	**Cellular RXD**	In	← SIM7080G TXD
5	**Cellular PWRKEY**	Out	Power control
6	**Cellular STATUS**	In	Module status
7	Buzzer	Out	PWM alert tone
8	Vibration motor	Out	On/Off
9-12	Servo PWM 1-4	Out	Gate control
13-16	Optical sensors 1-4	In	Pill counting
17	Button 1 (Confirm)	In	Pull-up
18	Button 2 (Cancel)	In	Pull-up
19	Battery voltage ADC	ADC	Fuel gauge backup
20-21	USB D-/D+	Bidir	Programming

Step 17.5 — Battery Installation

1. Connect Li-Po 3000mAh (JST-PH 2-pin)
2. **VERIFY POLARITY** (red=+, black=-)
3. Secure with foam tape (**NEVER screws near Li-Po**)
4. TP4056 LED should light when USB-C power connected

5 Full charge time: 3-4 hours at 1A

Step 17.6 — Travel Device Test Checklist

TEST	EXPECTED	PASS?
USB-C charging	LED lights, battery charges	☐
Fuel gauge (MAX17048)	Accurate battery %	☐
OLED display (SSD1306)	Shows 128×64 UI	☐
Cellular (LTE Cat-M1)	Network registration <30s	☐
GPS fix	Gets coordinates (1-3 min cold start)	☐
SIM card detection	SIM authenticated	☐
Servo 1-4	All open/close	☐
Optical sensors 1-4	All detect pills	☐
Buzzer	Beeps at set frequency	☐
Vibration motor	Vibrates	☐
API heartbeat via LTE	Sent and received by server	☐
Battery life (standby)	>14 days	☐
Battery life (4 doses/day)	>7 days	☐

PHASE 18

Production Testing & Quality Sign-Off

18.1 — In-Circuit Test (ICT)

TEST POINT	EXPECTED	MIN	MAX	MEASURED	PASS?
VIN (12V input)	12.0V	11.4V	12.6V	_____	☐
VSYS (BQ24195 out)	12.0V	11.0V	13.0V	_____	☐
5V rail	5.0V	4.75V	5.25V	_____	☐
3.3V rail	3.3V	3.2V	3.4V	_____	☐
Battery pack	7.4V	6.0V	8.4V	_____	☐
ESP32 VCC	3.3V	3.2V	3.4V	_____	☐

18.2 — Functional Test (5 min/unit)

STEP	TEST	PASS CRITERIA	PASS?
1	Power on	Boots in <10s	☐
2	WiFi connect	Connects to test AP	☐
3	Display test	All pixels OK, colors correct	☐
4	Touch test	4 corners + center	☐
5	Motor test (all 10)	All rotate smoothly	☐
6	Sensor test (all)	All respond correctly	☐
7	Audio test	1kHz tone at 80dB	☐
8	API heartbeat	Sent and received	☐
9	Serial number	Programmed (SMD-XXXXXXXX)	☐

18.3 — 24-Hour Burn-In

PHASE	DURATION	CONDITIONS	MONITORING
Normal	8 hours	25°C room temp	IC temps, power consumption
High temp	8 hours	40°C (climate chamber)	IC temps <60°C
Cycling	8 hours	Dispense every 10 min	Accuracy = 100%

18.4 — Quality Sign-Off Checklist

CHECKLIST ITEM	CHECK
All components verified	☐
PCB visual inspection (no bridges)	☐
All power tests passed	☐
All functional tests passed	☐
All calibrations saved	☐
Serial number programmed	☐
Firmware version: ___.___.___	☐
Enclosure sealed, labels applied	☐

18.5 — Sign-Off

ROLE	NAME	SIGNATURE	DATE
Electronics Engineer			
Firmware Engineer			
QA Inspector			

APPENDIX A

Master GPIO Reference — SMD-100

Source: Firmware `pins.h` (authoritative) cross-referenced with Technical Documentation Section 11.1

GPIO	FUNCTION	DIR	BUS/INTERFACE	NOTES
0	BOOT button / Confirm btn	In	—	10K pull-up
1-5	LCD R0-R4	Out	RGB parallel	Red channel (5 bits)
6-11	LCD G0-G5	Out	RGB parallel	Green channel (6 bits)
12-16	LCD B0-B4	Out	RGB parallel	Blue channel (5 bits)
15	I2S DOUT	Out	I2S	Audio data (shared with LCD_B3 — see note)
17	I2S BCLK	Out	I2S	Audio bit clock
18	I2S LRCLK	Out	I2S	Audio L/R clock
20	USB D-	Bidir	USB	USB OTG
21	USB D+ / LCD_HSYNC	Bidir/Out	USB / RGB	Dual function
26	PIR motion	In	Digital	AM312 output
27	Door/tray reed switch	In	Digital	10K pull-up
32	HX711 DOUT	In	Custom serial	Load cell data
33	HX711 SCK	Out	Custom serial	Load cell clock
34	Button Cancel	In	Digital	
35	SD CMD (MOSI)	Bidir	SPI	SD card
36	SD CLK	Out	SPI	SD card
37	SD DATA0 (MISO)	Bidir	SPI	SD card
38	Touch SDA	Bidir	**I2C Bus 0**	GT911 touch
39	Touch SCL	Out	**I2C Bus 0**	GT911 touch
40	Touch INT	In	Digital	Interrupt from GT911
41	Touch RST	Out	Digital	GT911 reset
42	Backlight PWM	Out	LEDC PWM	Via MOSFET to display
43	Gate servo PWM	Out	LEDC PWM	SG90 control
44	Sensor I2C SDA	Bidir	**I2C Bus 1**	SHT40, BH1750, MCP23017
45	LCD CLK	Out	RGB parallel	Pixel clock
46	Sensor I2C SCL	Out	**I2C Bus 1**	Sensor bus clock
47	LCD VSYNC	Out	RGB parallel	Vertical sync
48	LCD DE / LED data	Out	RGB parallel	Data enable
MCP23017 (I2C-Expanded GPIOs via Bus 1)				
GPA0-7	Motor control (motors 1-4)	Out	I2C-expanded	4 pins per motor
GPB0-7	Optical sensors 1-8 / Motors 5-8	In/Out	I2C-expanded	Interrupt-capable

Note on GPIO15: Firmware defines GPIO15 as I2S_DOUT, but it is also mapped as LCD_B3. If using RGB565 (16-bit) mode, only 5+6+5=16 data pins are needed. Depending on color depth,

GPIO15 may be available for audio. Verify in your specific schematic.

APPENDIX B
Power Consumption Budget

COMPONENT	ACTIVE CURRENT	SLEEP CURRENT	NOTES
ESP32-S3	240mA	10µA	WiFi TX peak
Display (4.3" TFT)	150mA	0mA	Backlight at 50%
Motors (1 active)	200mA	0mA	During dispense only
Sensors (all)	20mA	5mA	I2C polling
Audio (MAX98357A)	300mA	0mA	Peak during alert
Other (LEDs, misc)	30mA	5mA	
TOTAL	**940mA**	**20mA**	

Battery Life (5000mAh, 7.4V → 5V via buck):
Standby: 5000mAh / 20mA = **250 hours ≈ 10 days**
With 4 dispenses/day: **~48 hours** (meets target)

APPENDIX C
Common Pitfalls & Solutions

PITFALL	PROBLEM	SOLUTION
Motor EMI	Stepper motors cause WiFi interference	Add **100nF cap across each coil**, separate power paths, ferrite beads
WiFi interference	Motors running kills WiFi	Separate motor ground return, add LC filtering on motor power entry
Power brown-out	ESP32 resets when motor starts	Add **inrush current limiting** (soft-start), ensure supply handles 940mA peak
Touch erratic	Random touch events	**Ground display bezel**, use shielded FFC cable
Battery drain	Won't last 48 hours	Implement proper **deep sleep modes**, kill power to peripherals when idle
Load cell drift	Readings change over time	**Temperature compensation** (read SHT40 and correct), periodic tare
Audio distortion	Clipping at high volume	Limit gain to 12dB (GAIN pin), add soft clipping in firmware
Optical false counts	Ambient light triggers sensor	**Shield sensor** from room light, verify 100Ω LED resistor value

APPENDIX D

Troubleshooting Guide

Power Issues

SYMPTOM	CAUSE	FIX
No power	Barrel jack disconnected / blown fuse	Check connector continuity, replace fuse
5V rail missing	TPS62150 not soldered / inductor open	Reflow IC, check inductor continuity
3.3V rail low or absent	AP2112K bad joint / overloaded	Reflow, check load <600mA
BQ24195 overheating	Thermal pad not connected	Reflow — ensure thermal vias filled
Battery not charging	Wrong NTC resistor / BMS tripped	Check NTC value, check BMS fuse/MOSFETs

ESP32 Issues

SYMPTOM	CAUSE	FIX
Won't boot	EN pin floating	Add 10K pull-up to 3.3V
Boot loop	Flash corrupt or brownout	Re-flash firmware, check power stability
Stuck in download mode	GPIO0 held LOW	Check 10K pull-up on GPIO0
No serial output	Wrong baud / TX-RX swapped	Use 115200 baud, swap TX/RX wires
WiFi weak/disconnects	Metal near antenna area	Ensure 15mm clearance, no ground under antenna
PSRAM not detected	Wrong module variant	Verify marking shows "N16R8"

Display Issues

SYMPTOM	CAUSE	FIX
Blank screen	FFC not seated or reversed	Reseat cable, verify contact orientation
No backlight	2N7002 MOSFET dead / GPIO42 not driven	Replace MOSFET, check firmware PWM
Wrong colors	R/G/B channels swapped in firmware	Check `pins.h` RGB assignments vs FFC pinout
Touch not responding	I2C bus 0 not connected	Check GPIO38/39 wiring, 4.7K pull-ups
Touch erratic/jumpy	ESD damage or ungrounded bezel	Install TVS diodes, ground display frame

Motor Issues

SYMPTOM	CAUSE	FIX
Motor doesn't turn	ULN2003A dead	Replace driver IC
Turns wrong direction	Coil wires swapped	Fix order: Blue-Pink-Yellow-Orange
Motor stalls under load	5V not reaching motor	Check 5V at connector, ULN2003A COM to 5V
WiFi interference	No EMI filtering	Add 100nF across coils + ferrite on power

Sensor Issues

SYMPTOM	CAUSE	FIX
I2C device not found	Pull-ups missing / wrong address	Add 4.7K pull-ups, verify address pins
Load cell drifts	Temperature / mechanical stress	Compensate with SHT40, check mount
Optical miscounts	Ambient light leak	Shield sensor gap, increase LED current
PIR false triggers	Motor EMI / heat source nearby	Add 100nF filter cap, relocate away from motors

Debug Serial Commands

```
> status          System overview (WiFi, battery, temp, API status)
> motor test 1    Test motor 1 rotation
> motor test all  Test all motors sequentially
> sensor test     Read all sensors and display values
> dispense 1 2    Dispense from slot 1, 2 pills
> calibrate loadcell   Enter load cell calibration
> version         Show firmware version and build date
> i2c scan 0      Scan I2C bus 0 (touch)
> i2c scan 1      Scan I2C bus 1 (sensors)
> wifi status     Show WiFi connection details
> reboot          Restart device
```

APPENDIX E

Second-Source Strategy

Always have a backup supplier for critical components to avoid production delays.

CRITICAL COMPONENT	PRIMARY	SECOND SOURCE
MCU	ESP32-S3-WROOM-1-**N16R8**	ESP32-S3-WROOM-1-**N8R8** (less flash, still works)
Power Path IC	BQ24195	BQ25895 (newer, pin-compatible concept)
5V Buck	TPS62150	AP62150 (Diodes Inc, $0.80, 93% eff)
3.3V LDO	AP2112K	XC6220B331MR (lower Iq, $0.40)
Motor Driver	ULN2003AN (TI)	ULN2003A (ST) — identical spec
Temp Sensor	SHT40	SHT31 (slightly larger, $3.50)
Optical Sensor	TCPT1300X01	GP1A57HRJ00F (Sharp, 5mm gap, $1.20)
Load Cell ADC	HX711	NAU7802 (I2C, $1.50, smaller)

Contacts

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